

SQL Case Study

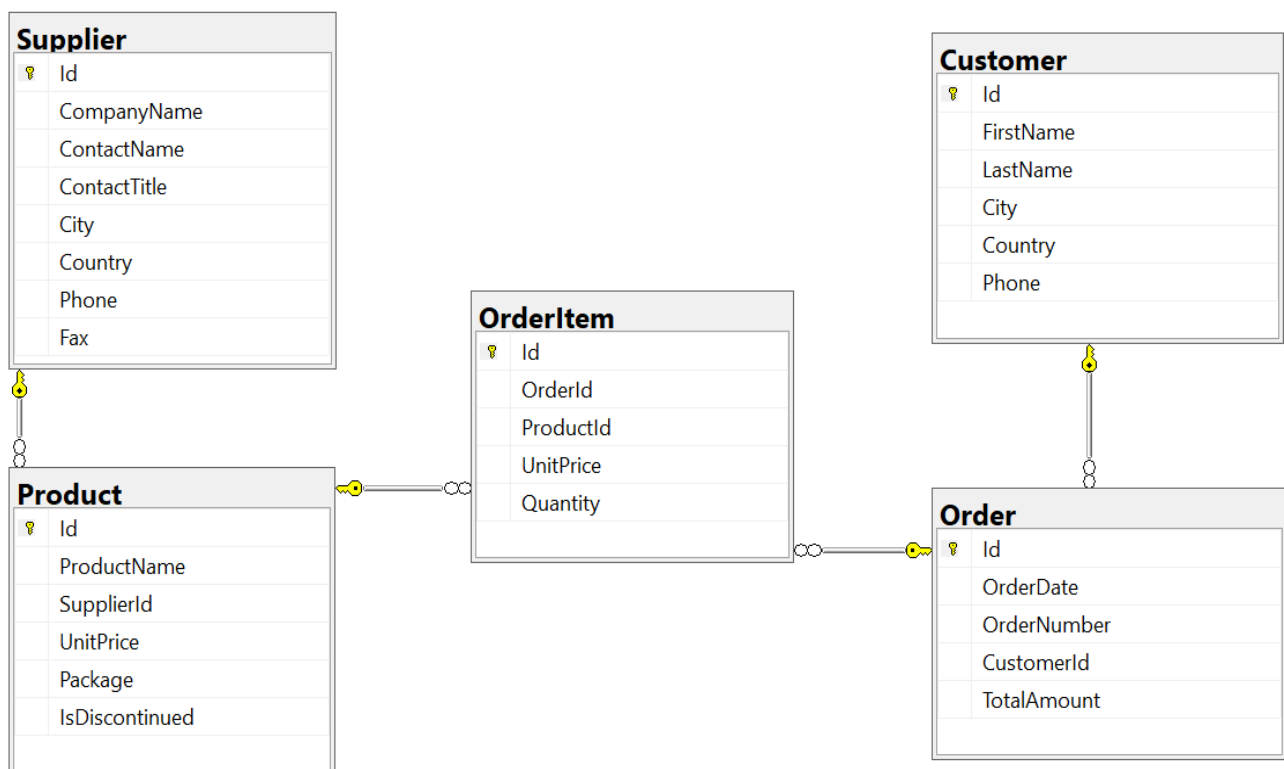
SQL Practice Case Study on SalesOrders Schema

- Our database is a modern version of Northwind traders.
- The data model consists of 5 tables.
- Supports SQL Server 2014, 2016, 2017, and 2019.
- Alternatively, use the live database in our SQL Editor.

Entity Relationship Diagram (ERD)

ERD stands for Entity Relationship Diagram. ERD provides a visual starting point for database design that can also be used to help determine information system requirements throughout an organization.

Below is the ERD for our sample database:



ERD diagrams are an important tool for data professionals: they provide a wealth of information, about tables, columns, data types, relationships, primary keys, foreign keys, and indexes -- all in a single diagram.

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Some Sample Basic Queries based on the above mentioned ERD:

1. List all suppliers in reverse alphabetical order.
2. List all customers in Italy.
3. List all customers in Paris sorted by last name.
4. Add Oxford Trading to the list of suppliers.
5. Supplier with id 24 has moved. Change their city from Sydney to Melbourne.
6. Remove 'Stroopwafels' from the products.
7. List all suppliers in the UK.
8. List all customers.
9. List the first name, last name, and city of all customers.
10. List all suppliers in France.
11. List all customers in Sweden.
12. For supplier Pavlova, Ltd, change the city to Sydney.
13. Delete products with a price higher than \$50.
14. An INSERT statement that adds a new supplier to the database.
15. Add new customer Craig Smith to the database.
16. The Bigfoot Brewery supplier has also become a customer. Add a customer record with values from the supplier table.
17. Change the phone number for supplier Tokyo Traders.
18. Discontinue all products in the database.
19. Discontinue all products over \$50.
20. Discontinue the product with Id = 46.

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Some Sample Advanced Queries based on the above mentioned ERD:

1. List the 10th to 15th most expensive products sorted by price.
2. List all French customer cities (without duplicates).
3. List all unique supplier countries in alphabetical order.
4. Find the number of unique supplier countries.
5. Find the largest order amount.
6. Find the cheapest products.
7. Find the largest order placed in 2014.
8. Find the last order date in 2013.
9. Get the total number of products.
10. Get the total number of customers.
11. Calculate the total sales in 2013.
12. Calculate the average size of all orders.
13. List all suppliers in Paris, France.
14. Get the customer named Thomas Hardy.
15. List all customers from Spain or France.
16. List all customers that are not from the USA.
17. List all orders that are not between \$50 and \$15000.
18. List the availability of all products.
19. List suppliers and the language they speak.
20. Categorize products as cheap, medium, or expensive.